

SIEMENS



ABT Site 6.0

Flexible ports topologies

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1 Cyber security disclaimer

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In order to protect plants, systems, machines and networks against cyber threats, it is necessary to implement – and continuously maintain – a holistic, state-of-the-art security concept. Siemens' portfolio only forms one element of such a concept.

You are responsible for preventing unauthorized access to your plants, systems, machines and networks which should only be connected to an enterprise network or the internet if and to the extent such a connection is necessary and only when appropriate security measures (e.g. firewalls and/or network segmentation) are in place. Additionally, Siemens' guidance on appropriate security measures should be taken into account. For additional information, please contact your Siemens sales representative or visit the following website:

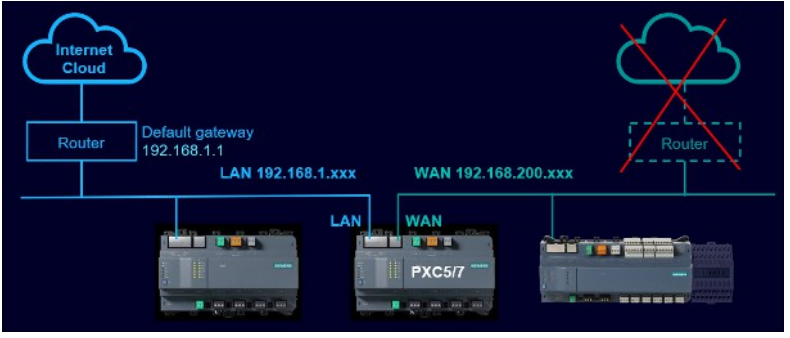
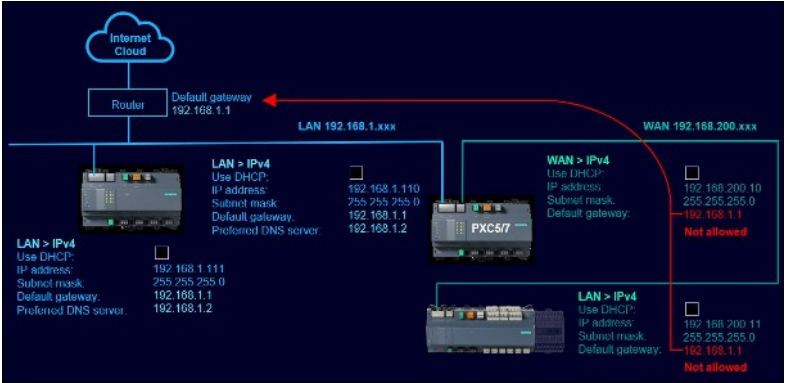
<https://www.siemens.com/global/en/products/automation/topic-areas/industrial-cybersecurity.html>.

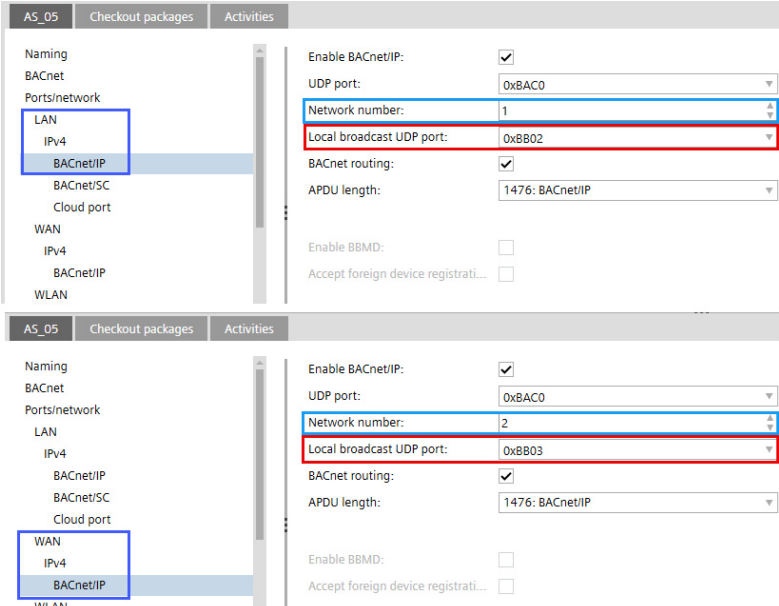
Siemens' portfolio undergoes continuous development to make it more secure. Siemens strongly recommends that updates are applied as soon as they are available and that the latest versions are used. Use of versions that are no longer supported, and failure to apply the latest updates may increase your exposure to cyber threats. Siemens strongly recommends to comply with security advisories on the latest security threats, patches and other related measures, published, among others, under the following website:

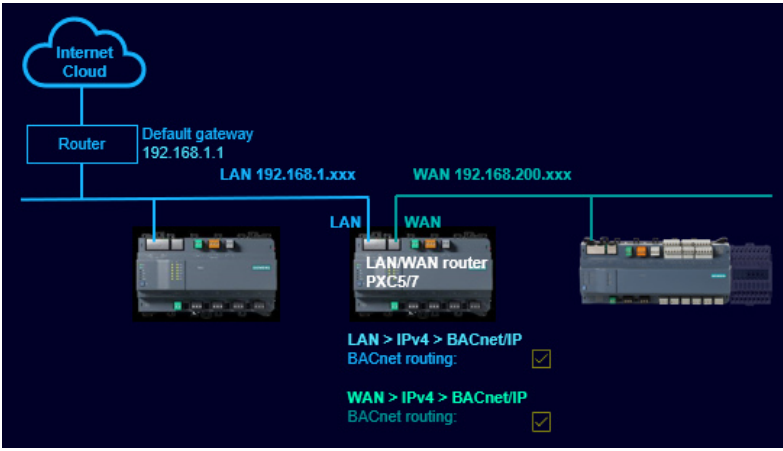
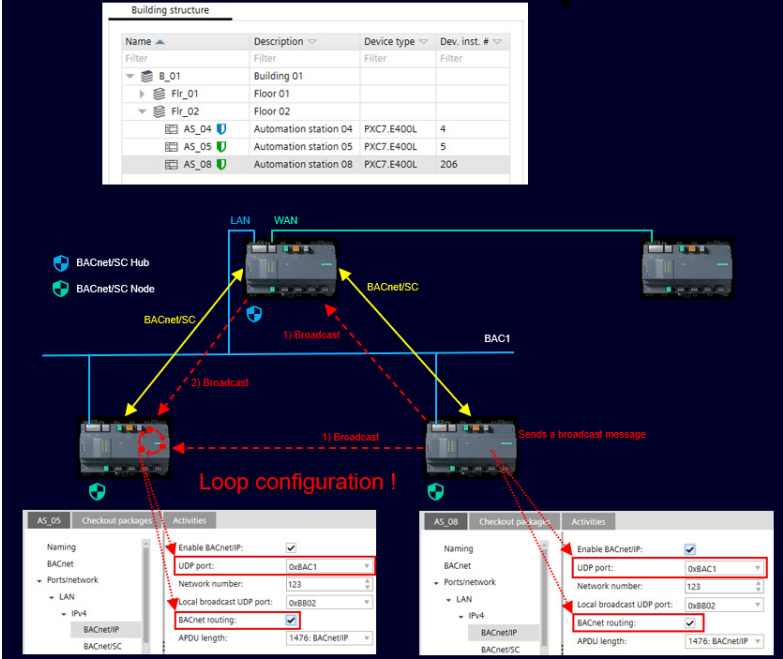
<https://www.siemens.com/cert/> => 'Siemens Security Advisories'.

2 Topology examples with settings and restrictions

The following information in the table are valid for all topologies. Specific information is mentioned separately for the corresponding topologies.

Topic	Description
Physical connection	All devices that will only communicate via one port must be connected via the LAN port.
Internet connection	A connection to the Internet is only possible from the LAN port. 
Default gateway	The connection to the Internet via the router (default gateway) must be made via the LAN port. A connection from the WAN port via a default gateway to the Internet is not supported (see also Internet connection above).
Default Gateway settings	It is not permitted to reference the default gateway from the WAN devices to the LAN router, for example, 192.168.1.1. The setting for all devices in the WAN subnet must be 0.0.0.0. 
DNS server	No DNS server is supported in the WAN subnet. DNS on the WAN side can only resolve WAN local addresses.

Topic	Description
Network numbers UDP broadcast ports configuration	- The network numbers on LAN and WAN side must be different. - The local broadcast UDP port of LAN port and WAN port must be different. 
BACnet routing	BACnet routing is only supported for BACnet communication, if BACnet routing is enabled on both sides. - No other traffic is routed between LAN subnet and WAN subnet. - Https traffic is not routed between the interfaces; routing web communication (https) between LAN and WAN is not supported. The Web UI can be accessed from either LAN or WAN subnet. Note: This means that no download can take place from LAN side to a device on WAN subnet, as this is not BACnet data.

Topic	Description
BACnet routing on LAN/WAN router	BACnet routing must be activated on the LAN/WAN router for the LAN port and the WAN port. BACnet routing maybe also switched on for MSTP connections.  Note: If several BACnet routers are routed to the same BACnet network, care must be taken to ensure that no loops are created. 
Download	The control program or firmware cannot be downloaded from the LAN port via the WAN port or vice versa. ABT Site must be connected to the respective subnet in order to perform a download (see also BACnet routing above). Topology with ABT Site [→ 12]
Engineering	- When a device is not configured, it cannot communicate using BACnet. Initial communication and setup of the device's BACnet configuration are necessary, allowing them to be discovered by tools like ABT or Web interface. - Until the bootstrapping of the devices for their initial engineering and commissioning is completed, discovery of devices through the device ports LAN->WAN or WAN->LAN is not possible. Thus, if devices on the WAN side are unconfigured, BACnet communication with them is not possible. You must first directly connect to those devices via WLAN, USB, or other WAN directly to discover, configure, or load them.

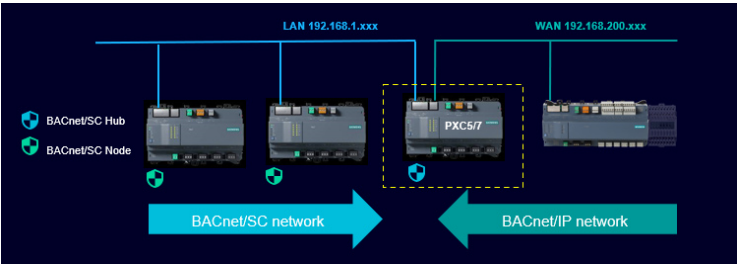
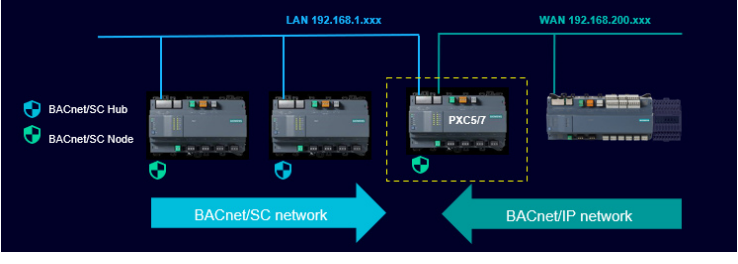
Topic	Description
Discovering devices	BACnet devices can be discovered on the opposite subnet using the Web UI: • Connect via HTTPS on LAN subnet to the Web server to discover devices on the WAN subnet. • Connect via HTTPS on WAN subnet to the built-in Web server to discover devices on the LAN subnet.
Commissioning	Engineering and commissioning from LAN to WAN and vice versa is not supported, but initial device configuration is always performed on-premise.

General information to the LAN / WAN topologies

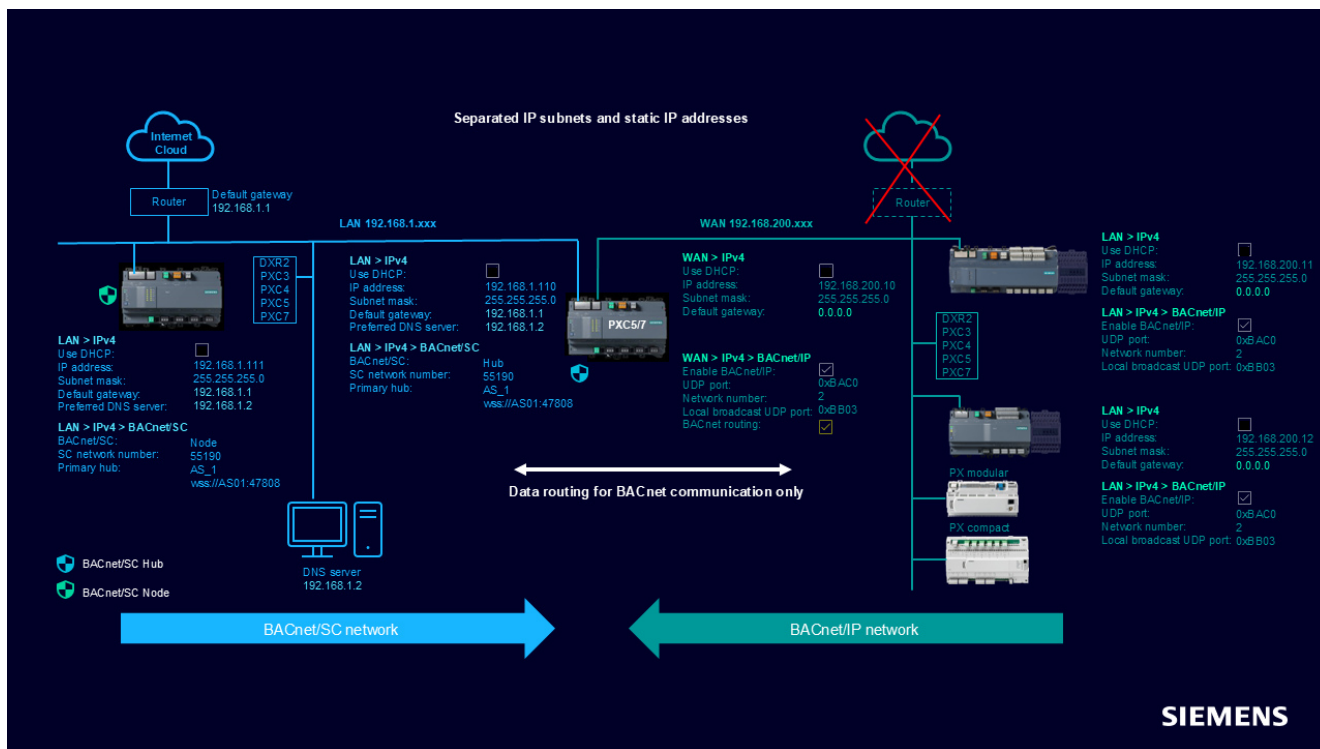
Recommendation
LAN and WAN are physically completely separated. Segments completely separated.
Due to the higher cybersecurity risk, IT departments may divide IT landscapes into distinct realms for office LAN and OT LAN(s), each with dedicated servers (DHCP, DNS).
Requires IT investment in strict separation and we expect this to happen in critical infra projects where you want to make a true separation on all levels.
Requires completely independent network infrastructures which must be created and maintained.
Connection over the Internet as an independent network is required.
On WAN side, we do not support multiple segments (no default gateway possible).

Not recommend scenario
LAN and WAN connected to the same segment - Two BACnet IP segments with different UDP ports, where all devices are on the same segment.

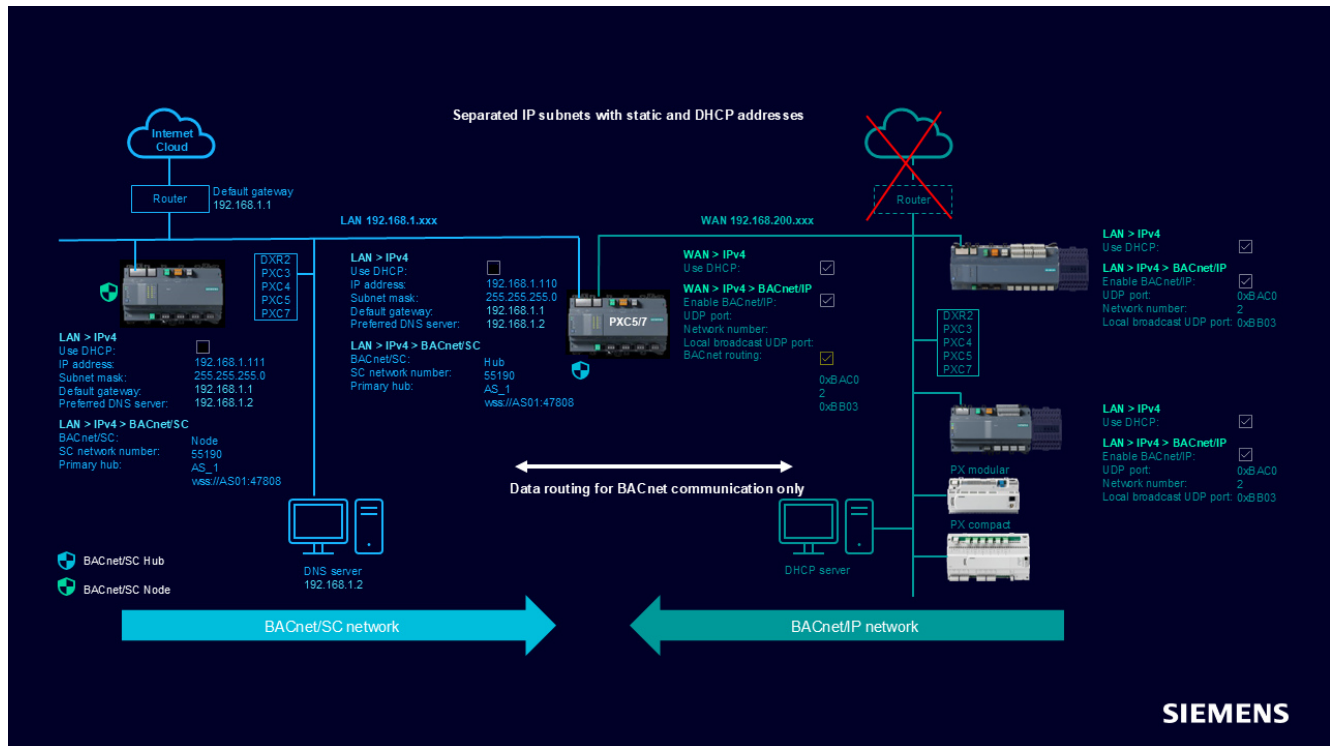
2.1 Topologies with BACnet/IP and BACnet/SC

Topic	Description
BACnet/SC device used as hub or node	- Hub: recommended for smaller systems with little communication LAN <-> WAN.  - Node: recommended for large systems.  - Node: recommended for heavily loaded LAN <-> WAN communication. Note: When configured as a node, the latency time for LAN <-> WAN communication increases by a few msec.
BACnet routing LAN/WAN device	With BACnet/SC, BACnet routing is automatically enabled on the LAN port. There is no checkbox in ABT Site to activate it. BACnet routing must be manually activated on the WAN port.

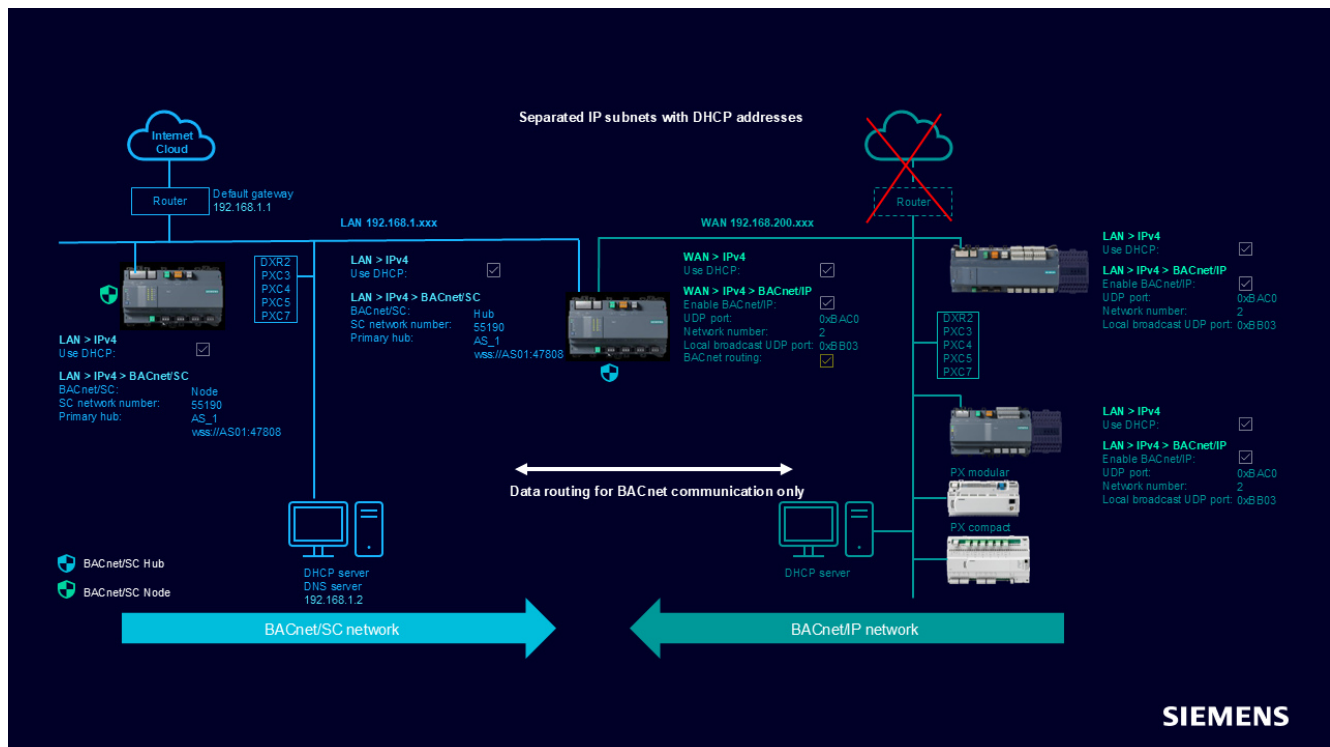
Static IP address on LAN subnet and WAN subnet



Static IP address on LAN subnet and DHCP address on WAN subnet



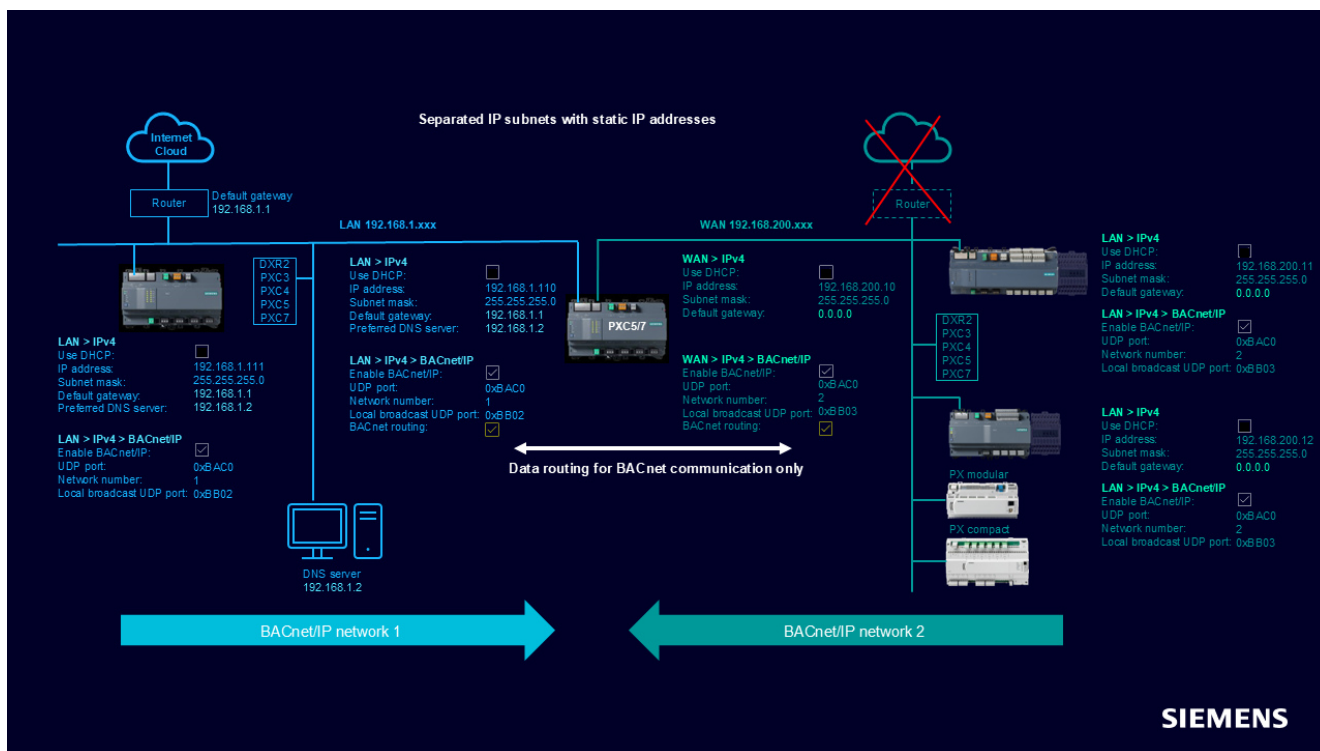
DHCP address on LAN subnet and on WAN subnet



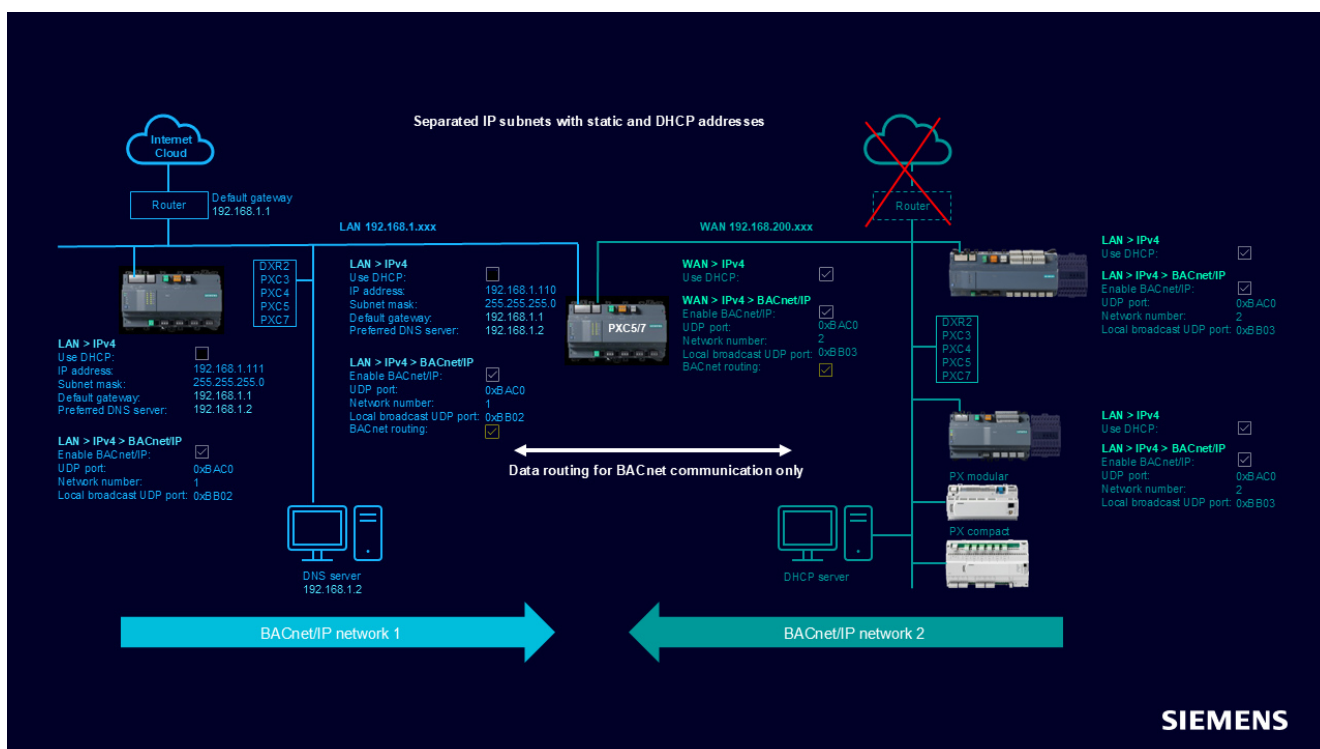
2.2 Topologies with BACnet/IP only

Topic	Description
BACnet routing LAN/WAN device	With separated IP subnets, the BACnet routing must be manually activated on the LAN port and the WAN port.

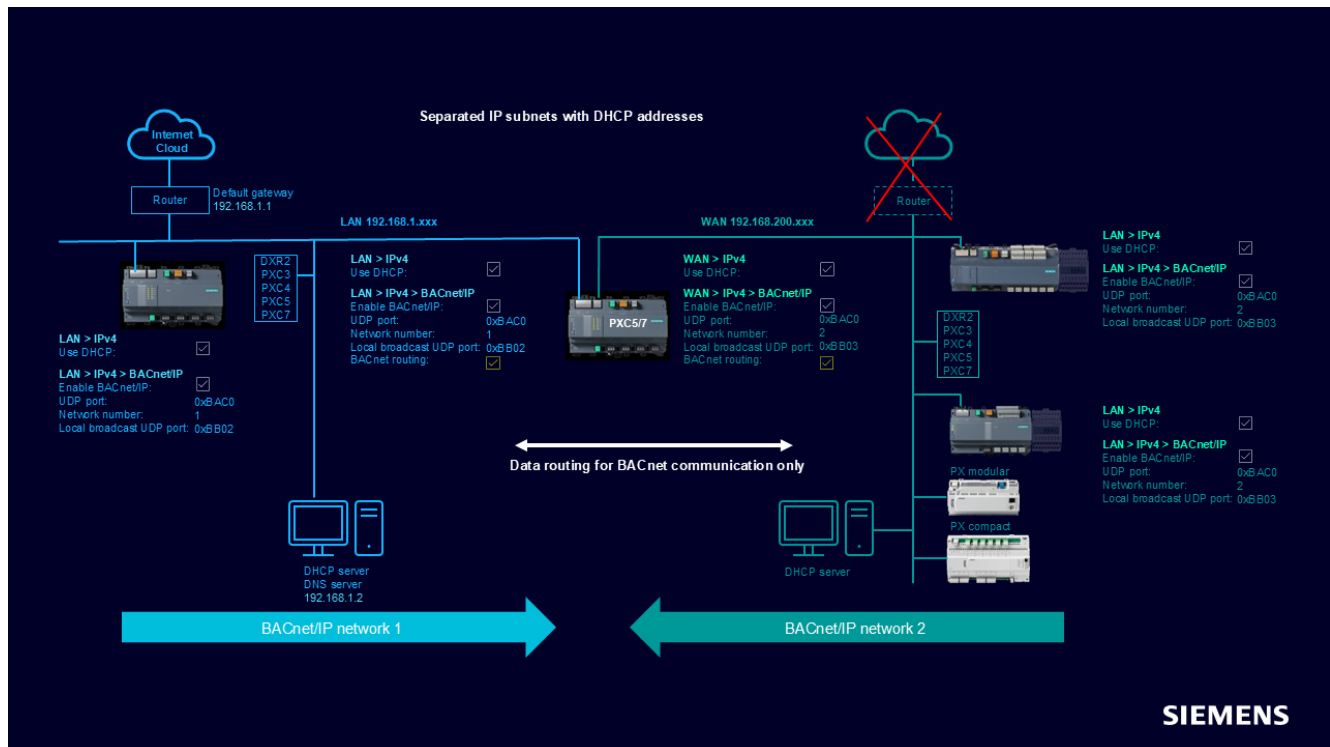
Static IP address on LAN subnet and WAN subnet



Static IP address on LAN subnet and DHCP address on WAN subnet



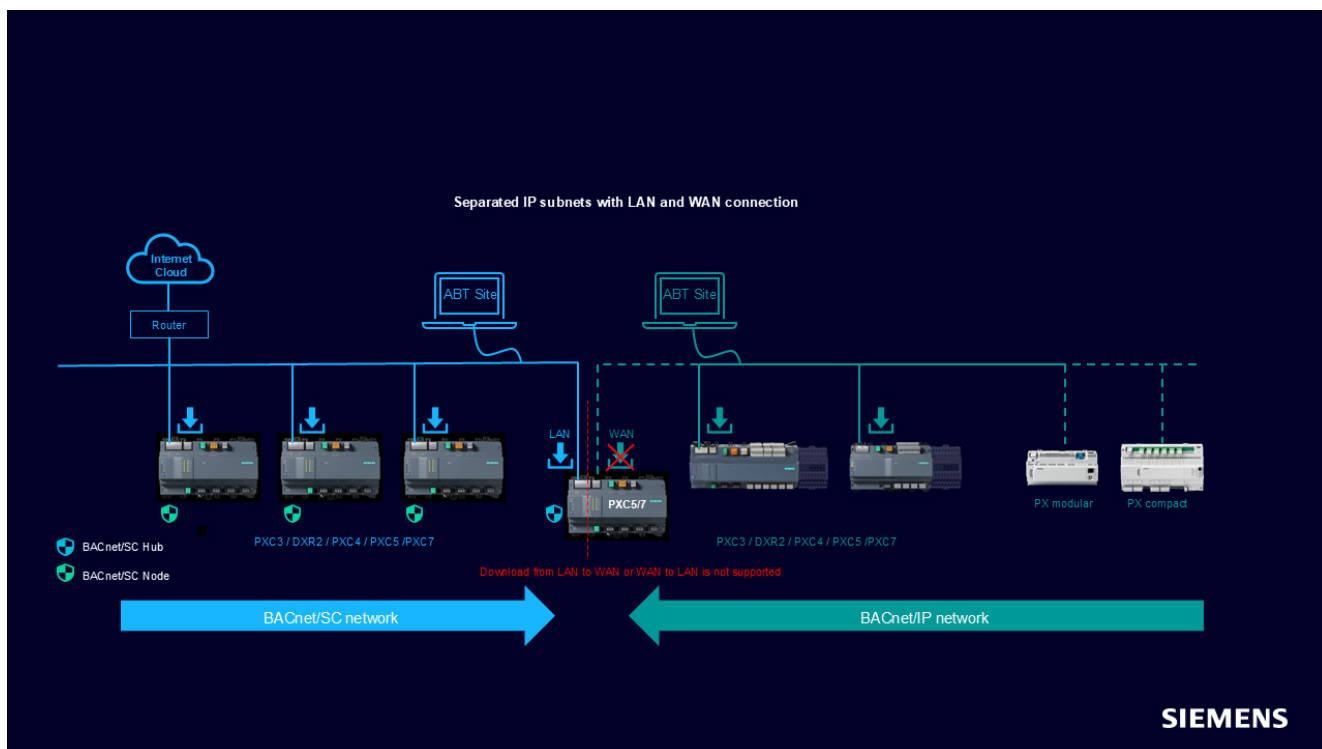
DHCP address on LAN subnet and on WAN subnet



2.3 Topology with ABT Site

Topic	Description
Download control program	The control program cannot be downloaded from the LAN port via the WAN port or vice versa. ABT Site must be connected to the respective subnet in order to perform a download.
Download firmware	The firmware cannot be downloaded from the LAN port via the WAN port or vice versa. ABT Site must be connected to the respective subnet in order to perform a download.
Download to LAN/WAN router	The control program or firmware must be downloaded via the LAN port. A download via the WAN port is not fully supported.

Connecting ABT Site on LAN port and/or WAN port



3 MAC addresses available for each port

MAC addresses are assigned at the factory and only apply to new devices starting with x date.

- 2 LAN ports, 1 WLAN port = 3 MAC addresses for the device types:
 - PXC4.E16-2, PXC4.E16S-2
 - PXC5.E003
- 2 LAN ports, 1 WAN port, 1 WLAN port = 4 MAC addresses for the device types:
 - PXC7.E400x
 - PXC5.E24

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PXC5.E24
 Variant: E24
 S55375-C104
 Country of Origin: Switzerland
 Date/Series: YYYYMMDD S0000001
 S/N: 140000A12F Hex
 KNX-ID: 00FD00xxxxxx Hex
 1st MACaddr: 000000000000 Hex
 No. of MACaddr: 4

PELV, Class 2
 AC 24 V~, 88 VA, 50/60 Hz
 DC 24 V=, 77 W
 -5 T 50, IP20 Indoor Use Only


 normally open:
 AC 12 V~.. 250 V~
 DC 12 V=.. 30 V=
 4 (3)A
 4 A Res, 3 A Gen. Purp.
 CAN ICES-3(B)/NMB-3(B)

normally closed:
 AC 12 V~.. 250 V~
 DC 12 V=.. 30 V=
 2 (1.5)A
 2 A Res, 1.5 A Gen. Purp.



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The device displays one MAC address with a QR code; the other MAC addresses are calculated by +1 (Hex).

Port	Device with 3 ports	Device with 4 ports
LAN 1	MAC 1 (QR-Code)	MAC 1 (QR-Code)
LAN 2	MAC 2	MAC 2
WAN		MAC 3
WLAN	MAC 3	MAC 4

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